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Multi-directional input device

Abstract

A multi-directional input device has: a housing; a tiltable manipulation member that swings around an axis; a linked member, which has a pivotally supported section supported by the housing so as to be swingable around the axis and also has a connecting portion in contact with the manipulation member, the linked member swinging in response to a tilting motion applied to the manipulation member; and a swing detection section that detects the swing of the linked member. The linked member has a flexible portion, located between a connecting portion and the pivotally supported section, that serves as a flexural fulcrum for the flexure of a first portion including the connecting portion with respect to a second portion including the pivotally supported section. The flexure of the first portion due to an external force applied to the connecting portion from the manipulation member is detected by a flexure detection unit.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a Continuation of International Application No. PCT/JP2023/004408 filed on Feb. 9, 2023, which claims benefit of Japanese Patent Application No. 2022-035032 filed on Mar. 8, 2022. The entire contents of each application noted above are hereby incorporated by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to a multi-directional input device that accepts an input made through a manipulation member when it is tilted in a desired direction.
2. Description of the Related Art
(2) Japanese Unexamined Patent Application Publication No. 11-024777 discloses a multi-directional switch that can be thinned and down-sized and needs only a small force to manipulate the multi-directional switch, as a multi-directional input device that accepts an input made through a manipulation member such as a manipulation lever when the manipulation member is tilted. In this multi-directional switch, a curved surface shape is formed so that as the manipulation lever is tilted, the point of a pressure contact between a movable member and the lower end of the manipulation lever moves toward the central axis of the manipulation lever.
(3) Japanese Unexamined Patent Application Publication No. 2000-305647 discloses a multi-directional input device that is superior in manipulation feeling for a manipulation shaft. In this multi-directional input device, the manipulation shaft and an actuating member are spline-coupled together. Therefore, when the manipulation shaft is rotated with it tilted, even if friction is generated between a bottom plate and the bottom of the actuating member due to a pressing force of an urging member, the actuating member, which is splined coupled to the manipulation shaft, is rotated together with the manipulation shaft. The actuating member rotates in such a way that it rolls, without slipping on the bottom plate.
(4) In a multi-directional input device, a linked member, the action of which is triggered by a tilting motion applied to the manipulation member, is provided. The linked member is pivotally supported by a housing, so the linked member swings around a predetermined swing axis with respect to the housing. Part of a detection portion, which detects the swing of the linked member, is attached to the linked member. When the detection portion is, for example, a magnetic detection portion, a magnet is disposed on the same side as the linked member and a magnetic sensor is disposed on the same side as a circuit board. When the linked member swings, a relative position between the magnet and the magnetic sensor changes. Therefore, a change of in magnetic force from the magnet is detected by the magnetic sensor to sense the swing of the linked member. Furthermore, when the manipulation member is pressed (the motion to press the manipulation member is referred to as a push motion), the linked member is displaced. A push sensor is activated due to this displacement to detect the push motion applied to the manipulation member. Since the linked member is involved in detecting both a tilting motion and a push motion applied to the manipulation member, it is necessary to prevent the push motion from affecting the detection of a swing motion.
SUMMARY OF THE INVENTION
(5) In view of the above situation, the present invention provides a multi-directional input device that can prevent a push motion applied to a manipulation member from affecting the detection of a

swing motion.

(6) A multi-directional input device in one aspect of the present invention has: a housing; a manipulation member, which is tiltable and swings around a first swing axis; a first linked member, which has a first pivotally supported section supported by the housing so as to be swingable around the first swing axis and also has a connecting portion in contact with the manipulation member, the first linked member swinging in response to a tilting motion applied to the manipulation member; and a first swing detection section that detects the swing of the first linked member. The first linked member has a first portion including the connecting portion, a second portion including the first pivotally supported section, and a flexible portion, located between the first portion and the second portion, that serves as a flexural fulcrum for the flexure of the first portion with respect to the second portion. The flexure of the first portion due to an external force applied to the connecting portion from the manipulation member is detected by a flexure detection unit.

(7) In this specification, the flexural fulcrum refers to a portion, of a member to be flexed, that includes the part at which the largest displacement will occur. Two portions (first portion and second portion) located on both sides of the flexural fulcrum undergo relative displacement substantially without their deformation. In this structure, the displacement of the first portion and the displacement of the second portion can be separated by the flexible portion, so the first pivotally supported section is less likely to be affected by the push motion applied to the manipulation member. This reduces the possibility that the push motion applied to the manipulation member is mistakenly detected as a tilting motion.

(8) In the above multi-directional input device, a portion eligible for detection by the flexure detection unit is preferably located further away from the flexible portion than the connecting portion is. Thus, the amount of displacement of the portion eligible for detection becomes larger than the amount of displacement of the connecting portion, making it easy to increase sensitivity in displacement detection.

(9) The multi-directional input device may further have an urging member that applies, to the manipulation member, a return force with which the manipulation member returns to a neutral position. The urging member may urge the manipulation member to press the first pivotally supported section of the first linked member against the housing, and when the external force is eliminated from the first linked member, may restore the flexed state of the first portion. Due to this type of urging force of the urging member, the manipulation member easily return to the neutral position and the flexed state of the first portion is easily restored.

(10) In the multi-directional input device, the housing preferably includes a holding portion that suppresses displacement of the second portion in a direction other than the direction of a swing around the first swing axis when an external force is applied to the connecting portion. Thus, the displacement of the second portion is suppressed by the holding portion when the first portion is flexed, so the first portion is flexed without the first pivotally supported section being affected.

(11) The above multi-directional input device may further have a second linked member, which has a second pivotally supported section supported by the housing so as to be swingable around a second swing axis crossing the first swing axis, the second linked member swinging in response to a tilting motion applied to the manipulation member, and may also have a second swing detection section that detects the swing of the second linked member. Thus, the manipulation member is tilted around two axes and its tilting is detected.

(12) In the multi-directional input device, the first swing detection section may have a magnetic force generation source disposed in the second portion and may also have a magnetic sensor disposed at a position at which a magnetic force from the magnetic force generation source can be measured. When the magnetic force generation source is disposed in the second portion, the adverse influence of the flexure of the first portion is reduced, enabling the swing of the manipulation member to be stably detected.

(13) In the above multi-directional input device, the bending rigidity of the flexible portion is

preferably lower than the bending rigidity of a first continuous portion connected to the flexible portion, the first continuous portion being part of the first portion, and is preferably lower than the bending rigidity of a second continuous portion connected to the flexible portion, the second continuous portion being part of the second portion. Thus, the first portion can be reliably flexed at the flexible portion.
(14) In the above multi-directional input device, the flexible portion, first continuous portion, and second continuous portion may have a portion integrally formed from the same material, and the flexible portion may have a portion thinner than either the first continuous part or the second continuous part. Thus, even when the flexible portion, first continuous portion, and second continuous portion are integrally formed from the same material, the flexible portion can be reliably flexed at the thinner portion.
(15) In the above multi-directional input device, a contact between the first pivotally supported section and the housing may be a rolling contact. In this case, the first swing axis may pass through a contact portion between the first pivotally supported section and the housing. In the above multi-directional input device, a rolling contact may be formed by a convex portion and a concave portion when viewed along the first swing axis. Due to this type of rolling contact, a frictional force at the contact portion becomes smaller than when the first pivotally supported section and housing make a sliding contact.
(16) The present invention can provide a multi-directional input device that can prevent a push motion applied to a manipulation member from affecting the detection of a swing motion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating a multi-directional input device according to an embodiment;
(2) FIG. **2** is also a perspective view illustrating the multi-directional input device according to the embodiment;
(3) FIG. **3** is an exploded perspective view illustrating the structure of the multi-directional input device according to the embodiment;
(4) FIG. **4** is a perspective view illustrating a tilting motion applied to a manipulation member;
(5) FIG. **5** is a sectional view of the multi-directional input device according to the embodiment;
(6) FIG. **6** is a schematic view to explain the action of a first linked member;
(7) FIG. **7** is also a schematic view to explain the action of the first linked member;
(8) FIG. **8**A is a sectional view to explain a push motion applied to the manipulation member;
(9) FIG. **8**B is also a sectional view to explain a push manipulation by the manipulation member;
(10) FIG. **9**A is also a sectional view to explain a push motion applied to the manipulation member;
(11) FIG. **9**B is also a sectional view to explain a push motion applied to the manipulation member;
(12) FIG. **10**A is also a sectional view to explain a push motion applied to the manipulation member; and
(13) FIG. **10**B is also a sectional view to explain a push motion applied to the manipulation member.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
(14) An embodiment of the present invention will be described below in detail with reference to the accompanying drawings. In the descriptions below, like members will be denoted by like reference characters and repeated descriptions will be appropriately omitted for members that have been described once.
(15) Structure of a Multi-Directional Input Device
(16) FIGS. **1** and **2** are each a perspective view illustrating a multi-directional input device

according to this embodiment.

(17) FIG. **3** is an exploded perspective view illustrating the structure of the multi-directional input device according to this embodiment.

(18) The multi-directional input device **1** according to this embodiment accepts an input made through a manipulation member **20** when it is tilted with respect to a housing **10**.

(19) In the description of the embodiment, it will be assumed that a first swing axis AX**1**, which is a first of the swing axes involved in the tilting motion applied to the manipulation member **20**, is parallel to the X axis, and a second swing axis AX**2**, which is a second of the swing axes, is parallel to the Y axis, and that the axis (neutral axis AX**3**) of the manipulation member **20** at a neutral position is parallel to the Z-axis.

(20) The multi-directional input device **1** has the housing **10**, the manipulation member **20**, a first linked member **30**, a second linked member **40**, an urging member **50**, a first swing detection section **60**, a second swing detection section **70**, and a displacement detection section **80**. The housing **10** is in a substantially box shape with an opening at the bottom. A hole **10***h*, in which the manipulation member **20** is placed, is formed at the center of the upper portion of the housing **10**. A bottom plate member **15** is attached to the opening at the bottom of the housing **10**. A frame plate member **17** is attached to a side surface of the housing **10**. The bottom plate member **15** may be formed as part of the housing **10**. Non-limiting examples of the constituent materials of the housing **10** and bottom plate member **15** include metal materials such as iron-based materials, aluminum-based materials, and copper-based materials. The bottom plate member **15** may be formed from a material different from that of the housing **10** (for example, a resin-based material such as polyester (such as polybutylene terephthalate) or polyamide).

(21) The manipulation member **20** has a cylindrical portion **21** placed in the interior of the housing **10**, and also has an extending portion **22** that extends from the interior of the housing **10** through the hole **10***h* to the outside. When the manipulation member **20** is at the neutral position, an extending direction D, in which the extending portion **22** extends, is parallel to the Z-axis. When the manipulation member **20** is tilted, the extending direction D of the extending portion **22** is non parallel to the Z-axis. The manipulation member **20** can be tilted around the first swing axis AX**1** and second swing axis AX**2**, with respect to the housing **10**.

(22) The first linked member **30** has a first pivotally supported section **31**, which is supported by the housing **10** so as to be swingable around the first swing axis AX**1**, the first linked member **30** swinging in response to the tilting motion applied to the manipulation member **20**. In the housing **10**, a pressure receiving portion **11** is provided that is pressed by the first pivotally supported section **31**. The first linked member **30** is in a frame shape having a hole **30***h* at the center. The manipulation member **20** is inserted into the hole **30***h* at the center of the first linked member **30**. A fitting protrusion **23** protrudes from the cylindrical portion **21** of the manipulation member **20**. The fitting protrusion **23** is slidably fitted into a fitting hole **30***a* formed in the first linked member **30**. Non-limiting examples of the constituent material of the first linked member **30** include resin-based materials such as polyacetal, polyester (such as polybutylene terephthalate), and polyamide. Details of the first linked member **30** will be described later.

(23) The second linked member **40** may have a second pivotally supported section **41**, which is supported by the housing **10** so as to be swingable around the second swing axis AX**2**, the second linked member **40** swinging in response to the tilting motion applied to the manipulation member **20**. In the housing **10**, a pivotally supported contact **12** is provided so as to be in contact with the second pivotally supported section **41**. The second linked member **40** has an arch portion **42** curved in an arch shape. A hole **42***h* is formed at the center of the arch portion **42** of the second linked member **40**. The extending portion **22** of the manipulation member **20** is inserted into the hole **42***h* at the center of the arch portion **42** of the second linked member **40**. The extending portion **22** of the manipulation member **20** has a convex portion **22***a*. When the manipulation member **20** is inserted into the hole **42***h* in the arch portion **42**, the convex portion **22***a* comes into contact with

the arch portion **42**, and the extending portion **22** is slidably fitted into the hole **42***h*.
(24) The second linked member **40** is placed so as to pass over the first linked member **30** in the Y-axis direction. In a state in which the second linked member **40** passes over the first linked member **30** and the extension portion **22** of the manipulation member **20** is inserted into the hole **30***h* in the first linked member **30** and into the hole **42***h* in the second linked member **40**, the second linked member **40**, first linked member **30**, and manipulation member **20** are incorporated into the interior of the housing **10**.
(25) The urging member **50** may urge the manipulation member **20** to press the first pivotally supported section **31** of the first linked member **30** against the housing **10**, and may also apply, to the manipulation member **20**, a return force with which the manipulation member **20** returns to the neutral position. The urging member **50** is, for example, a coil spring. The urging member **50** is inserted into the cylindrical portion **21** of the manipulation member **20**. There is a bottom cover **25** at the bottom of the cylindrical portion **21**, into which the urging member **50** has been inserted. The bottom cover **25** is provided so as to be slidable within the cylindrical portion **21** in an extending direction D, in which the extending portion **22** extends. The bottom cover **25** is in contact with the bottom plate member **15**. Thus, the urging member **50** is sandwiched between the bottom cover **25** and an inner upper wall **21***a* (see FIG. **5**) of the cylindrical portion **21**. The urging member **50** gives an urging force to the manipulation member **20**.
(26) When the manipulation member **20** is tilted, the bottom cover **25** in contact with the bottom plate member **15** receives a reaction force from the bottom plate member **15** and thereby slides along the extending direction D, pressing the urging member **50**. When the tilting motion applied to the manipulation member **20** is canceled, the manipulation member **20** returns to the neutral position due to the urging force of the urging member **50**.
(27) The first swing detection section **60** detects the swing of the first linked member **30**. The second swing detection section **70** may detect the swing of the second linked member **40**. The first swing detection section **60** may have, for example, a magnetic sensor **61** and a permanent magnet (magnet **62**), which is a magnetic force generation source. The second swing detection section **70** has, for example, a magnetic sensor **71** and a permanent magnet (magnet **72**). The magnetic sensors **61** and **71** are mounted on a circuit board **90**. The circuit board **90**, on which the magnetic sensors **61** and **71** are mounted, is placed on the bottom plate member **15**.
(28) The magnet **62**, which faces the magnetic sensor **61**, is placed in a pocket **30***p* formed in the first linked member **30**. The magnet **72**, which faces the magnetic sensor **71**, is placed in a pocket **40***p* formed in the second linked member **40**. The magnet **62** swings around the first swing axis AX**1** when the first linked member **30** swings. The magnet **72** swings around the second swing axis AX**2** when the second linked member **40** swings. Due to the swings of the magnets **62** and **72**, their positions relative to the magnetic sensors **61** and **71** fixed to the circuit board **90** change. The resulting changes in magnetic field strength are detected by the magnetic sensors **61** and **71**. The swings of the first linked member **30** and the second linked member **40** are detected by signals output from the magnetic sensors **61** and **71**.
(29) The bottom plate member **15** has a component mounting portion **15***a*, which extends laterally. The displacement detection section **80** is attached to the component attachment section **15***a*. The displacement detection section **80** is, for example, a switch of contact detection type such as a Tact switch (registered trademark). The displacement detection unit **80** detects displacement of the manipulation member **20** in a direction different from both a direction around the first swing axis AX**1** and a direction around the second swing axis AX**2**. In this embodiment, the displacement detection section **80** detects displacement along the direction in which the manipulation member **20** extends.
(30) The first linked member **30** has an arm **33**, which extends above the displacement detection portion **80**, from a side opposite to the side on which the first pivotally supported section **31** is disposed. For example, when the manipulation member **20** is pushed in a direction opposite to the

direction in which the manipulation member **20** extends from the housing **10** (this pressing motion is also referred to as a push motion), a fulcrum is located on the same side as the first pivotally supported section **31**, and the arm **33** of the first linked member **30** is pressed toward the displacement detection portion **80** due to the pressing force. Thus, the arm **33** comes into contact with the displacement detection section **80**, activating the displacement detection section **80**. The push motion will be described later in detail.
(31) Tilting Motion Applied to the Manipulation Member
(32) FIG. **4** is a perspective view illustrating a tilting motion applied to the manipulation member **20**.
(33) [P**0**] in FIG. **4** illustrates a state in which the manipulation member **20** is at the neutral position. That is, when no manipulation force is applied to the manipulation member **20**, in which case there is no load, the manipulation member **20** is at the neutral position as illustrated at [P**0**]. In this embodiment, when the manipulation member **20** is at the neutral position, the extending portion **22** of the manipulation member **20** extends along a direction intersecting both the first swing axis AX**1** and the second swing axis AX**2**, specifically the Z-axis direction.
(34) When a manipulation force is applied to the manipulation member **20** in the direction of arrow a with the manipulation member **20** at the neutral position, the first linked member **30** swings around the first swing axis AX**1** and the manipulation member **20** tilts as illustrated at [P**1**] in FIG. **4**. When the manipulation force applied to the manipulation member **20** is eliminated in the state at [P**1**], the manipulation member **20** returns to the neutral position at [P**0**] due to the urging force of the urging member **50**. Conversely, when a manipulation force is applied to the manipulation member **20** in the direction of arrow b with the manipulation member **20** at the neutral position, the first linked member **30** swings around the first swing axis AX**1** in a direction opposite to the direction at [P**1**] and the manipulation member **20** tilts as illustrated at [P**2**] in FIG. **4**. When the manipulation force applied to the manipulation member **20** is eliminated in the state at [P**2**], the manipulation member **20** returns to the neutral position at [P**0**] due to the urging force of the urging member **50**.
(35) When a manipulation force is applied to the manipulation member **20** in the direction of arrow c with the manipulation member **20** at the neutral position, the second linked member **40** swings around the second swing axis AX**2**, and the manipulation member **20** swings as illustrated at [P**3**] in FIG. **4**. When the manipulation force applied to the manipulation member **20** is eliminated in the state at [P**3**], the manipulation member **20** returns to the neutral position at [P**0**] due to the urging force of the urging member **50**. Conversely, when a manipulation force is applied to the manipulation member **20** in the direction of the arrow d with the manipulation member **20** at the neutral position, the second linked member **40** swings around the second swing axis AX**2** in a direction opposite to the direction at [P**3**] and the manipulation member **20** tilts as illustrated at [P**4**] in FIG. **4**. When the manipulation force applied to the manipulation member **20** is eliminated in the state at [P**4**], the manipulation member **20** returns to the neutral position at [P**0**] due to the urging force of the urging member **50**.
(36) When a manipulation force is applied to the manipulation member **20** in a direction other than the directions of arrows a, b, c, and d with the manipulation member **20** at the neutral position, the first linked member **30** and the second linked member **40** swing according to components in the directions of arrows a, b, c, and d in the direction in which the manipulation force is applied and the manipulation member **20** tilts to a position other than at [P**1**], [P**2**], [P**3**], and [P**4**]. That is, the manipulation member **20** can tilt in a direction at any angle within 360 degrees when viewed along the Z-axis.
(37) First Linked Member
(38) FIG. **5** is a sectional view of the multi-directional input device **1** according to this embodiment.
(39) The sectional view in FIG. **5** is taken along a plane that includes the first swing axis AX**1** and

is orthogonal to the Y-axis.
(40) FIGS. **6** and **7** are schematic views to explain the motion applied to the first linked member **30**.
(41) At the first pivotally supported section **31**, the first linked member **30** is pivotally supported by the housing **10**. The structure in this pivotal support is not limited. In this embodiment, as an example, the contact between the first pivotally supported section **31** and the housing **10** may be a rolling contact, and the first swing axis AX**1** may pass through the contact portion between the first pivotally supported section **31** and the housing **10**. In a specific example of this structure, a rolling contact may be formed at the contact portion between the first pivotally supported section **31** and the housing **10** by a convex portion and a concave portion when viewed along the first swing axis AX**1**.
(42) In this embodiment, the housing **10** has a pressure receiving portion **11**, which is pressed by the first pivotally supported section **31**. The pressure receiving portion **11** is, for example, a V-shaped protrusion. The first pivotally supported section **31** has a V-shaped recess for receiving the V-shaped protrusion of the pressure receiving portion **11**. When viewed along the first swing axis AX**1**, the recessed portion in the first pivotally supported section **31** includes a V-shape wider than the angle of the V-shaped protrusion of the pressure receiving portion **11**. When the V-shaped protrusion of the pressure-receiving portion **11** and the V-shaped valley of the recess in the first pivotally supported section **31** come into contact with each other, a rolling contact is made around the first swing axis AX**1**. Due to this type of rolling contact, a frictional force at the contact portion becomes smaller than when the first pivotally supported section **31** and housing **10** make a sliding contact.
(43) The first pivotally supported section **31** is provided only on one side of the first linked member **30** on the first swing axis AX**1**. The arm **33** is disposed on a side opposite to the first pivotally supported section **31** of the first linked member **30** on the first swing axis AX**1**.
(44) When the fitting protrusion **23** fits into the fitting hole **30***a* formed in the first linked member **30**, the manipulation member **20** is linked to the first linked member **30**. The fitting hole **30***a*, into which the fitting protrusion **23** is fitted, is included in a connecting portion **305** in the first linked member **30**. On the first swing axis AX**1**, therefore, the first linked member **30** is supported at the position on the fitting protrusion **23** of the manipulation member **20** and at the position on the first pivotally supported section **31** on one side with respect to the position of the manipulation member **20**.
(45) In this embodiment, the first linked member **30** has a first portion **301**, a second portion **302**, and a flexible portion **300**. The first portion **301** includes the connecting portion **305**, and the second portion **302** includes the first pivotally supported section **31**. The flexible portion **300** is located between the first portion **301** and the second portion **302**, and serves as a flexural fulcrum for the flexure of the first portion **301** with respect to the second portion **302**.
(46) As illustrated in FIG. **6**, while the manipulation member **20** undergoes no push motion, an urging force is applied from the urging member **50** to the manipulation member **20**, urging the first linked member **30** in the extending direction D of the extending portion **22**. In this state, therefore, the first portion **301** of the first linked member **30** is not flexed, so the first portion **301** and second portion **302** are not substantially displaced relative to each other. Since the first portion **301** is not flexed, a gap t between the displacement detection section **80** and the arm **33** of the first linked member **30** is maintained, preventing the displacement detection section **80** from being activated.
(47) As illustrated in FIG. **7**, when a push motion is applied to the manipulation member **20**, an external force F is applied to the manipulation member **20** in a direction opposite to the extending direction D. Since the manipulation member **20** is connected to the first linked member **30** through the connecting portion **305**, the external force F is transmitted to the first linked member **30** through the connecting portion **305**. When the external force F is transmitted to the connecting portion **305**, the first portion **301** of the first linked member **30** is flexed with respect to the second

portion **302** with the flexible portion **300** serving as a flexural fulcrum. Due to this flexure, the first portion **301**, which includes the connecting portion **305**, and the second portion **302** are displaced relative to each other. Specifically, the first portion **301** is displaced in the direction opposite to the extending direction D.

(48) The arm **33** of the first linked member **30** may be located further away from the flexible portion **300** than the connecting portion **305** is. Therefore, a relationship is established in which the flexible portion **300** serves as a fulcrum, the connecting portion **305** serves as a point of effort, and the arm **33** serves as a point of action. Therefore, the amount of displacement of the arm **33** in the direction opposite to the extending direction D is larger than the amount of displacement of the connecting portion **305**, so the arm **33** comes into contact with the displacement detection portion **80** and activates it. In FIG. **7**, this displacement of the arm **33** is indicated by the clockwise arrow. In this way, the displacement detection section **80** functions as a flexure detection unit that detects the flexure, of the first portion **301**, which is caused by the external force F generated by a push motion. Thus, the push motion is detected.

(49) When the first linked member **30** receives the external force F from the manipulation member **20** due to a push motion, the first portion **301** is flexed with the flexible portion **300** serving as a flexural fulcrum. Even upon the receipt of the external force F by the first linked member **30**, therefore, the second portion **302** is less likely to be affected by the displacement of the first portion **301** including the connecting portion **305**. That is, since the flexible portion **300** is disposed, the displacement of the first portion **301** and the displacement of the second portion **302** can be separated. Therefore, the first pivotally supported section **31** included in the second portion **302** is less likely to be affected by the push motion applied to the manipulation member **20**.

(50) When the first linked member **30** receives an external force F′, which causes the first linked member **30** to undergo displacement (to swing) around the first swing axis AX**1**, from the manipulation member **20**, the first pivotally supported section **31** undergoes displacement (swings) around the first swing axis AX**1** according to the external force F′. In this way, the first linked member **30** can act differently depending on the direction of the external force that the first linked member **30** receives from the manipulation member **20**. Thereby, the multi-directional input device **1** according to this embodiment can appropriately identify the external force F and the external force F′.

(51) The magnet **62** facing the magnetic sensor **61** is placed in the second portion **302**. Even when a push motion is applied to the manipulation member **20**, the second portion **302** is not easily affected by the push motion. Therefore, a change in the relative position between the magnet **62** and the magnetic sensor **61** is suppressed when the push motion is performed. This reduces the possibility that the push motion applied to the manipulation member **20** is mistakenly detected as a tilting motion.

(52) When a push motion is applied to the manipulation member **20** and an external force F is thereby applied to the connecting portion **305** of the first linked member **30**, the second portion **302** may be displaced in a direction other than the direction of a swing around the first swing axis AX**1**. Therefore, it is preferable to provide a holding portion **307** to suppress this displacement. The holding portion **307** is disposed, for example, in the housing **10** or on the bottom plate member **15**. The holding portion **307** is placed so as to face a portion **302***a*, which is part of the second portion **302**, on a side opposite to the first portion **301** with respect to the first swing axis AX**1**.

(53) This portion **302***a* is easily affected by the displacement (displacement due to flexure or return) of the first portion **301** when the external force F is applied or eliminated. However, with the holding portion **307** placed so as to face this portion **302***a*, when the external force F is applied to the connecting portion **305** of the first linked member **30** and the first portion **301** is thereby displaced, the displacement of the second portion **302** can be suppressed. Specifically, it is possible to restrain the second portion **302** from being displaced clockwise together with the first portion **301**. This reduces the possibility that the first pivotally supported section **31** is similarly displaced

and the first swing axis AX**1** is offset.
(54) During the use of the multi-directional input device **1**, an external force F with which a push motion is applied and an external force F′ with which a swing around the first swing axis AX**1** occurs may be applied simultaneously. Even in this case, the offset of the first swing axis AX**1** due to the external force F is appropriately suppressed. Therefore, the multi-directional input device **1** can detect the displacement due to the external force F and can appropriately control the swing due to the external force F′.
(55) In the structure as described above, in which the first portion **301** of the first linked member **30** is flexed relative to the second portion **302** with the flexible portion **300** serving as a flexural fulcrum, the bending rigidity of the flexible portion **300** is preferably lower than the bending rigidity of a first continuous portion connected to the flexible portion **300**, the first continuous portion being part of the first portion **301**, and also lower than the bending rigidity of a second continuous portion connected to the flexible portion **300**, the second continuous portion being part of the second portion **302**. Thus, when the first portion **301** is flexed, the flexible portion **300** can be reliably flexed and deformed and the deformation of other portions can be suppressed. Therefore, the adverse influence particularly on the detection of the swing of the second portion **302** can be further reduced.
(56) When the flexible portion **300**, first continuous part, and second continuous part are integrally formed from the same material, it suffices for the flexible portion **300** to have a portion thinner than either the first continuous part or the second continuous part. Thus, even when the flexible portion **300**, first continuous portion, and second continuous portion are integrally formed from the same material, the bending rigidity of the thin portion of the flexible portion **300** can be made lower than the bending rigidity of the first continuous portion and the bending rigidity of second continuous portion, so flexure at the flexible portion **300** becomes possible.
(57) As described above, in this embodiment, to make the flexure of the first portion **301** more likely to be detected, the arm **33**, which is a portion eligible for detection by the displacement detection section **80**, may be located further away from the flexible portion **300** than the connecting portion **305** is. Thus, the amount of displacement of the arm **33** becomes larger than the amount of displacement of the connecting portion **305**, making it easy to increase sensitivity in displacement detection. For example, even if the push stroke of a push motion is short, the displacement detection section **80** can be reliably activated.
(58) Push Motion Applied to the Manipulation Member
(59) FIGS. **8**A to **10**B are each a sectional view to explain a push motion applied to the manipulation member **20**.
(60) Each sectional view in FIGS. **8**A to **10**B is taken along a plane that includes the first swing axis AX**1** and is orthogonal to the Y-axis.
(61) FIG. **8**A illustrates a state in which the manipulation member **20** is at the neutral position, and FIG. **8**B illustrates a state in which a push motion is applied to the manipulation member **20** at the neutral position.
(62) As illustrated in FIG. **8**A, before the manipulation member **20** at the neutral position undergoes a push motion, an urging force is applied from the urging member **50** to the manipulation member **20**, urging the first linked member **30** in the extending direction D of the extending portion **22**. In this state, although the arm **33** of the first linked member **30** is in contact with the displacement detection section **80**, the contact is not enough to activate the displacement detection section **80**.
(63) As illustrated in FIG. **8**B, when a push motion is applied to the manipulation member **20**, the manipulation member **20** is pushed against the urging force from the urging member **50**. Thus, the same side as the arm **33** of the first linked member **30** that is fitted to the manipulation member **20** is flexed with the flexible portion **300** of the first linked member **30** serving as a flexural fulcrum, and is pressed toward the displacement detection section **80**.

(64) The first linked member **30**, fitted to the manipulation member **20**, is supported at the position of the first pivotally supported section **31** on the first swing axis AX**1**. Therefore, the fitting position between the manipulation member **20** and the first linked member **30** (specifically, the fitting position between the fitting hole **30***a* and the fitting protrusion **23** illustrated in FIG. **3**) is closer to the first pivotally supported section **31** than to the arm **33** in the direction along the first swing axis AX**1**. Thus, when a pressing force due to the push motion applied to the manipulation member **20** is applied to the first linked member **30**, a fulcrum (flexural fulcrum) is located on the same side as the flexible portion **300** of the first pivotally supported section **31**, the fitting position between the manipulation member **20** and the first linked member **30** serves as a point of effort, and the tip of the arm part **33** serves as a point of load. Therefore, the amount of displacement caused by the push motion applied to the manipulation member **20** is magnified at the arm **33**, and the arm **33** reliably has a contact with the displacement detection section **80** located below the arm **33**, activating the displacement detection section **80**.
(65) FIG. **9**A illustrates a state in which the manipulation member **20** is tilted around the second swing axis AX**2**, and FIG. **9**B illustrates a state in which a push motion is applied to the manipulation member **20** at a tilted position.
(66) As illustrated in FIG. **9**A, when the manipulation member **20** is tilted around the second swing axis AX**2**, the bottom cover **25** is also tilted together with the manipulation member **20**. Thus, the bottom cover **25** receives a reaction force from the bottom plate member **15** and is pressed in the extending direction D, acting to contract the urging member **50** in the cylindrical portion **21**. In this state, although the arm **33** of the first linked member **30** is in contact with the displacement detection section **80**, the contact is not enough to activate the displacement detection section **80** because the urging force of the urging member **50** is applied to the first linked member **30**.
(67) As illustrated in FIG. **9**B, when a push motion is applied to the manipulation member **20** in the tilted state, the manipulation member **20** is pushed against the urging force from the urging member **50**. Thus, the same side as the arm **33** of the first linked member **30** fitted to the manipulation member **20** is flexed with the flexible portion **300** of the first linked member **30** serving as a flexural fulcrum, and is pressed toward the displacement detection section **80**. As a result, the arm **33** in contact with the displacement detection section **80** activates the displacement detection section **80**.
(68) FIG. **10**A illustrates a state in which the manipulation member **20** is tilted around the first swing axis AX**1**, and FIG. **10**B illustrates a state in which a push motion is applied to the manipulation member **20** at a tilted position.
(69) As illustrated in FIG. **10**A, when the manipulation member **20** is tilted around the first swing axis AX**1**, the first linked member **30** also swings together with the manipulation member **20**. Along with the tilt of the manipulation member **20**, the bottom cover **25** also tilts. Thus, the bottom cover **25** receives a reaction force from the bottom plate member **15** and is pressed in the extending direction D, acting to contract the urging member **50** in the cylindrical portion **21**. Although the first linked member **30** swings and the arm **33** of the first linked member **30** is in contact with the displacement detection section **80**, the contact is not enough to activate the displacement detection section **80** because the urging force of the urging member **50** is applied to the first linked member **30**.
(70) As illustrated in FIG. **10**B, when a push motion is applied to the manipulation member **20** in the tilted state, the manipulation member **20** is pushed against the urging force from the urging member **50**. Thus, the same side as the arm **33** of the first linked member **30** fitted to the manipulation member **20** is flexed with the flexible portion **300** of the first linked member **30** serving as a flexural fulcrum, and is pressed toward the displacement detection section **80**. As a result, the arm **33** in contact with the displacement detection section **80** activates the displacement detection section **80**.
(71) As described above, the multi-directional input device **1** according to this embodiment can

prevent a push motion applied to the manipulation member **20** from affecting the detection of a swing motion.

(72) This completes the description of the embodiment of the present invention. However, the present invention is not limited to examples in the embodiment. For example, the multi-directional input device **1** may have a structure that lacks the second linked member **40** and second swing detection section **70**. The scope of the present invention also includes embodiments obtained as a result of appropriately adding or deleting constituent elements to or from the above embodiment, performing design changes to the above embodiment, or appropriately combining features in the examples of the structure in the above embodiment without departing from the intended scope of the present invention; the addition, deletion, design change, or combination is effected by a person having ordinary skill in the art.

Claims

1. A multi-directional input device comprising: a housing; a manipulation member, which is tiltable and swings around a first swing axis; a first linked member, which has a first pivotally supported section supported by the housing so as to be swingable around the first swing axis and also has a connecting portion in contact with the manipulation member, the first linked member swinging in response to a tilting motion applied to the manipulation member; a first swing detection section that detects a swing of the first linked member; and a flexure detection unit; wherein the first linked member has a first portion including the connecting portion, a second portion including the first pivotally supported section, and a flexible portion, located between the first portion and the second portion, that serves as a flexural fulcrum for a flexure of the first portion with respect to the second portion, and the flexure of the first portion due to an external force applied to the connecting portion from the manipulation member is detected by the flexure detection unit.
2. The multi-directional input device according to claim 1, wherein a portion eligible for detection by the flexure detection unit is located further away from the flexible portion than the connecting portion is.
3. The multi-directional input device according to claim 1, further comprising an urging member that applies, to the manipulation member, a return force with which the manipulation member returns to a neutral position, wherein the urging member urges the manipulation member to press the first pivotally supported section of the first linked member against the housing, and when the external force is eliminated from the first linked member, restores the flexed state of the first portion.
4. The multi-directional input device according to claim 1, wherein the housing includes a holding portion that suppresses displacement of the second portion in a direction other than a direction of a swing around the first swing axis when the external force is applied to the connecting portion.
5. The multi-directional input device according to claim 1, further comprising: a second linked member, which has a second pivotally supported section supported by the housing so as to be swingable around a second swing axis crossing the first swing axis, the second linked member swinging in response to a tilting motion applied to the manipulation member; and a second swing detection section that detects a swing of the second linked member.
6. The multi-directional input device according to claim 1, wherein the first swing detection section has a magnetic force generation source disposed in the second portion and also has a magnetic sensor disposed at a position at which a magnetic force from the magnetic force generation source is measurable.
7. The multi-directional input device according to claim 1, wherein bending rigidity of the flexible portion is lower than bending rigidity of a first continuous portion connected to the flexible portion, the first continuous portion being part of the first portion, and is lower than bending rigidity of a second continuous portion connected to the flexible portion, the second continuous portion being

part of the second portion.
8. The multi-directional input device according to claim 7, wherein the flexible portion, the first continuous portion, and the second continuous portion have a portion integrally formed from the same material, and the flexible portion has a portion thinner than either the first continuous part or the second continuous part.
9. The multi-directional input device according to claim 1, wherein a contact between the first pivotally supported section and the housing is a rolling contact, and the first swing axis passes through a contact portion between the first pivotally supported section and the housing.
10. The multi-directional input device according to claim 9, wherein the rolling contact is formed by a convex portion and a concave portion when viewed along the first swing axis.